

Ahmad Samim Hamza

AI Engineer | Production AI Systems | Software Engineer

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ABOUT ME

AI/ML Engineer and Master's student in Artificial Intelligence at the University of Jyväskylä, with 5+ years of experience building production systems across backend, full-stack, and AI domains. Currently designing and deploying enterprise RAG and agentic applications at Konecranes using AWS Bedrock, Bedrock AgentCore, LangChain, and LangGraph — integrating LLMs with real business data and workflows. I combine solid software engineering foundations (Python, FastAPI, PostgreSQL, AWS, CI/CD) with deep applied AI expertise across conversational UIs, multi-agent orchestration, and vector retrieval systems. I work natively with AI-assisted coding tools (Claude Code, Cursor) and actively follow the fast-moving agentic AI landscape, prototyping with new frameworks as they emerge.

SKILLS

AI & GenAI: RAG architectures, embeddings, vector search, LangChain, LangGraph, AWS Bedrock (Agents, AgentCore, Knowledge Bases, MCP/tools), ChromaDB, prompt engineering, multi-agent orchestration, conversational UI design

AI-Assisted Coding: Claude Code, Cursor — daily use for accelerating full-stack and AI feature development

Machine Learning: TensorFlow/Keras, PyTorch, scikit-learn, NumPy, Pandas, Matplotlib, feature engineering, hyperparameter tuning

Backend & APIs: Python (FastAPI, Django), PHP (Laravel), Node.js/TypeScript (NestJS, Express), REST, auth & RBAC

Frontend: React, Next.js, Vue.js, Nuxt.js, Livewire, Alpine.js

Cloud & DevOps: AWS (Bedrock, Amplify, ECS, RDS, S3, Lambda, API Gateway), PostgreSQL, MySQL, Docker, Linux, Git, CI/CD, logging & monitoring

WORK EXPERIENCE

Summer Trainee, Software Development Specialist (AI/ML), Konecranes 05/2026 – Present
Hyvinkää, Finland

- Built multiple production-ready RAG applications on AWS Bedrock using Agents and AgentCore, over internal XML, PDF, and XLSX enterprise documents indexed in an S3-based vector store.
- Implemented RAG variants with AWS Bedrock, Bedrock AgentCore, LangChain and LangGraph, integrated into a unified Next.js + FastAPI application exposing different RAG backends behind a shared conversational UI.
- Developed the entire Next.js + FastAPI application using **Claude Code** as the primary AI-assisted coding environment, accelerating iteration speed while maintaining production-grade code quality.
- Architected a multi-agent system on AWS Bedrock with a supervisor agent orchestrating multiple collaborator agents, enabling modular, task-specific agentic workflows across enterprise knowledge sources.
- Used PostgreSQL for user accounts, session data, and feedback storage, enabling evaluation and iteration across RAG variants.
- Deployed the system using AWS Amplify, ECS, and RDS; collaborated with stakeholders to align chatbot capabilities and agentic workflows with business requirements.

SENIOR SOFTWARE ENGINEER, RAMAKI GROUP OF COMPANIES 05/2024 – 07/2025
KABUL, AFGHANISTAN

- Led and coordinated a team of 8 engineers in the development of a large-scale e-commerce platform.

- Designed and optimized PostgreSQL database schemas for complex transactional data.
- Built RESTful APIs; wrote unit, integration, and feature tests to ensure data integrity.
- Conducted code reviews and mentored junior developers, fostering a collaborative development environment.

SOFTWARE ENGINEER, PEACE GLOBAL LOGISTICS — 09/2022 – 04/2024 — KABUL, AFGHANISTAN

- Developed supply chain, logistics, and shipment tracking modules using NestJS, React, and Next.js.
- Designed and optimized PostgreSQL databases for logistics operations, handling order, inventory, and shipment data.
- Implemented reusable UI components in React + Tailwind for logistics dashboards.

Kabul University, Instructor (Python & Machine Learning) — 04/2022 – 10/2022 — Kabul, Afghanistan

- Taught Python programming and Machine Learning fundamentals to freshman and sophomore students.
- Covered data analysis, model training and evaluation, and practical ML tooling.

FULL STACK DEVELOPER, SMART FRIQI — 07/2021 – 09/2022 — KABUL, AFGHANISTAN

- Built ERP modules using Laravel, Nuxt.js, and PostgreSQL; added real-time features using WebSockets; developed dashboards for business management and operations monitoring.

PROJECTS

Enterprise RAG Assistant (Konecranes)
AWS Bedrock (Agents, AgentCore), LangChain, LangGraph, FastAPI, Next.js, PostgreSQL, AWS Amplify/ECS/RDS

Internal enterprise knowledge assistant over heterogeneous documents (XML, PDF, XLSX), with multiple RAG variants (Bedrock-native, LangChain, LangGraph), a shared conversational UI, and a feedback-based evaluation pipeline. Fully built using Claude Code as the primary development environment.

Health Information RAG Chatbot — MSc Mini Project, University of Jyväskylä (2026)
Python, LangChain, ChromaDB, FastAPI, Selenium, BeautifulSoup

Built a production-grade domain-specific RAG chatbot by scraping terveyskyla.fi using Selenium and BeautifulSoup, processing content into a ChromaDB vector store, and integrating an LLM for conversational health information retrieval. Full pipeline: web scraping → document parsing → vector ingestion → conversational retrieval interface.

Other Engineering Projects (2021–2024)

PGL ERP System (NestJS, React, PostgreSQL) · United Used Cars (Laravel, Next.js, MySQL) · TeebAlhoor ERP (Laravel, Nuxt.js, PostgreSQL) — logistics, supply chain, and ERP modules across full-stack roles.

EDUCATION

University of Jyväskylä, Master of Science in Artificial Intelligence — 08/2025 – Present — Jyväskylä, Finland

Kabul University, Bachelor of Computer Science (Software Engineering) — 2017 – 2021 — Kabul, Afghanistan

LANGUAGE SKILLS

PERSIAN ●●●●● ENGLISH ●●●●●

Finnish ●○○○○